



FABRIC V8.3.0 RELEASE NOTES

These Release Notes describe the new features in Fabric release V8.3.0 and list bugs that have been fixed since the latest V8.2 release.

Certification of this Fabric release is based on:

- Cassandra version 4.1.3
- SQLite version 3.46.1
- OpenJDK Runtime Environment 21.0.3
- Confluent Kafka version 7.6
- Neo4j 5.25.1 - enterprise
- Elasticsearch - 8.5.3
- AWS OpenSearch – 2.17
- PostgreSQL 15.4

MAIN FEATURES AND IMPROVEMENTS

1. Fabric Catalog

The Discovery & Catalog solution has been enhanced with several powerful features, such as:

- ✓ The ability to create a Catalog from files or HTTP requests.
- ✓ The ability to create a bulk of Catalog's nodes and edit this bulk (e.g. add new property to all bulk entries at once).
- ✓ Additional Catalog plugins with new capabilities, such as the creation of logical relations between datasets, analysis of source data and calculation of various metrics.

Below is a detailed list of the new features:

- **File Cataloging:** Catalog can now be built based on files or HTTP requests. Discovery can be performed either by the file's metadata definition or by sample data or both.
 - The Crawler framework used for file cataloging employs a generic mechanism that is independent of specific file format. It expects to get an input in a predefined format, achieved through the invocation of the 3 pre-defined Broadway flows: *Get Metadata*, *Get Files List*, *Get File Data*. These flows should be created in the project's implementation and attached to the project interface.
 - The *Files Cataloging - Demo* extension (on K2exchange) provides an illustration of the E2E process of file cataloging and masking the files data. The extension includes the flows examples for CSV, XML, JSON, AVRO and HTTP formats.
 - Two dedicated actors are provided to transform the JSON or AVRO schema format into Catalog metadata format.

https://support.k2view.com/Academy/articles/39_fabric_catalog/05_cataloging_of_files.html



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- **Bulk creation & edit:** The Catalog Application now supports a new manual activity: bulk creation and edit of the Catalog entities. This feature allows users to efficiently update multiple Catalog entities at once, by adding or editing various properties simultaneously (for example, assigning a *Sequence Name* property to multiple fields).
 - Bulk creation is initiated by selecting the required Catalog entities (such as fields) from the Search Catalog screen. The selected bulk is stored in the session storage and remains available until the browser session is closed.
 - Bulk editing is available only when the Catalog is in Edit mode. However, bulk creation and viewing are possible in any Catalog mode.

https://support.k2view.com/Academy/articles/39_fabric_catalog/catalog_app/14_1_bulk_creation.html
https://support.k2view.com/Academy/articles/39_fabric_catalog/catalog_app/14_2_bulk_edit.html
- **Logical relations plugins** were enhanced to provide more ways of identification of possible logical relations between the datasets.
 - **Reference by Query Analysis** is a new plugin that analyzes a list of SQL queries provided as an input file. By examining the JOIN operations in the queries, the plugin can create foreign key (FK) relationships between datasets. The query parsing and analysis is performed using either an internal Fabric parser or an LLM.
 - **Reference by Data Comparison** is a new plugin that examines data within the data source fields to identify correlations (using the probabilistic algorithm Bloom filter). Based on the analysis results, the plugin can establish FK relationships between datasets.
 - The existing plugin, formerly known as Metadata Logical Reference, has been renamed to **Reference by Names Comparison**. The rule for creating links between two non-PK fields across datasets has been removed.

https://support.k2view.com/Academy/articles/39_fabric_catalog/plugins/05_relations_creation.html
- **Source data analysis plugins** were enhanced to provide more statistical and analytical data that can be used to improve the quality of masking and synthetic data generation:
 - **Data Quality Metrics** is an existing plugin that has been enhanced with additional statistical metrics calculations, such as min, max, average and standard deviation.
 - **Option Set Analyzer** is a new plugin that identifies fields with a limited number of distinct values in a field (in data sample). The identified distinct values are then saved into a *catalog_field_option_set___<dataPlatform>_<schema>* MTable.

https://support.k2view.com/Academy/articles/39_fabric_catalog/plugins/04_source_data_metrics.html
- **Manual override of the relations** has been enhanced with the following:
 - It is now possible to create cross-schema / cross-data platform relations manually (when the Catalog is in Edit mode).
 - It is now allowed to create a relation from any field of the parent table, and not necessarily from the primary key.

https://support.k2view.com/Academy/articles/39_fabric_catalog/catalog_app/07_manual_overrides.html



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- **Catalog artifact** has been enhanced with the following:
 - The artifacts of the **relations** can now be created, when it is required. This feature is only available via the `/api/catalog/{version}/build-catalog-artifacts` API, by setting the input parameter `refersTo=true`. The relations artifacts are created as a new **catalog_relations_info** MTable.
 - The artifacts (both fields and relations) are now **split by default** into separate files, per data platform & schema, to reduce the time of each artifact's creation. This is done due to setting `SPLIT_CATALOG_ARTIFACTS = ON` (in config.ini) to become the default value.

https://support.k2view.com/Academy/articles/39_fabric_catalog/catalog_app/09_build_artifacts.html

https://support.k2view.com/Academy/articles/39_fabric_catalog/20_catalog_APIs.html

- Databases with 2 or 4 levels of hierarchy:
 - The Catalog hierarchy has 3 levels: data platform, schema and dataset.
 - Some databases might either have no schema level or have an additional hierarchy level (called a catalog or a database) that can group several schemas.
 - Catalog supports these data sources in the following way:
 - When schema level is missing or empty (e.g. SpannerDB), a schema is created with the name “__default” (2 underscore). Note that for SQLite databases, the Catalog keeps “main” as a schema name.
 - When there is an additional hierarchy level (e.g. Snowflake DB), it is reflected by a new property **catalogName**, created on a schema node.

2. Native Support for NoSQL Document Storage

Fabric can now provide native E2E support for NoSQL Document Storage (such as MongoDB or CouchBase), including the following:

- Discovery job can run on an interface of MongoDB or CouchBase instance, once the respective K2exchange connector has been installed in the project. Then the Catalog is created based on the discovered document hierarchy. (This feature is already supported prior to 8.3)
- The Web Studio's Interface explorer can now present the Document's complex structures such as nested hierarchy levels and arrays of primitives.
- The Logical Unit can now be created based on the Document's metadata retrieved from the Catalog, after discovery has run on it.
 - The nested hierarchy levels are then created as LU tables with a referential link to their respective parent level.
 - When a complex structure on any level has a non-unique name, the names of the parent levels are concatenated to it, after 3 underscores. E.g. `emailInfo__emergencyContacts`.
- The LU tables are created with the following system-generated fields:



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- `_docId` is a unique ID added to each LU table. Its purpose is to uniquely identify the row of an instance, when splitting the document and composing it back.
- `_parentDocId` is added to all LU tables except for the root table, and it is used for creating the referential link from the nested structure to its parent structure.
- `_value` is only added to LU tables that represent an array of primitives, to keep the element's value.
- `_docHints` is a field used internally by Fabric to manage the composition of the original Document.
- The root table population is the only population that reads data from the source (the Document).
- The populations created for such LU tables include a dedicated DocumentQuery actor that is responsible for generating unique IDs to maintain the relations between hierarchy levels of the Document. In addition, the actor takes the respective part of the original Document and breaks it into the fields, populated in the LU table.
- The TDM 9.4 templates have been enhanced to support LUs, generated based on NoSQL documents and enable loading the masked documents to the target environment:
 - Adding the TDM setup to these LUs.
 - The TDM delete flow is enhanced to delete these documents from the target system.
 - The TDM load flow is enhanced to assemble the document structure of a LU representing a document (complex type) using the DocumentAssemble actor and load the document to the target environment.

3. Broadway

New Actors

- **JsonSchemaToMetadata**, **AvroSchemaToMetadata** are new actors that can be used to transform the Json schema / Avro schema file into the Catalog metadata format. Their response is an array of maps (objects). These actors can be used when creating the "Get Metadata" flow as part of **File Cataloging** solution.

Existing Actors

- The **CatalogMasking** actors default mapping behavior has been updated as follows:
 - If the `catalog_field_info` MTable doesn't exist, the value is passed 'as is' without masking.
 - If the `catalog_field_info` MTable exists but the field is missing, its value is passed 'as is' without masking.

4. Web Studio

- The Interface Explorer tab displays databases with 2 or 4 hierarchy levels as follows:



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- When schema level is missing or empty (e.g. SpannerDB), a dummy schema name “__default” (2 underscore) is displayed. Note that for SQLite, “main” is displayed.
- When there is an additional hierarchy level (e.g. Snowflake DB), it is only reflected by a tooltip over schema node, displaying the Catalog Name.
- A new command has been added to the Web Studio and is available via the Command Palette. Its purpose is to reset the Git credentials:
 - "Git - Reset Saved Credentials Data"

5. Miscellaneous

- **BATCH command** has been enhanced with the support of **binding parameters**. You can utilize it by setting up one of the following new input parameters:
 - **BIND_PARAMS** - list of binding parameters for the SELECT query on the source interface.
 - **BIND_PARAMS_SEPARATOR** – separator between parameters in BIND_PARAMS (optional). The default separator is ','.
 - **BIND_PARAMS_JSON** – JSON-format array with binding parameters for the SELECT query on the source interface.
- **SET IS_STUDIO** is a new Fabric command to establish if the Fabric instance is Dev (includes Studio) or None-Dev (without Studio). The command is read only and returns true or false.
- **LIST CONFIG_OVERRIDES**, **LIST CONFIG_OVERRIDES_HISTORY** are new Fabric commands to list the user overrides of the Fabric configuration.
 - **LIST CONFIG_OVERRIDES** presents a list of current overrides.
 - **LIST CONFIG_OVERRIDES_HISTORY** presents the history of overrides, including the version, date of change, user, node id and the new value.

RESOLVED ISSUES

- Ticket #42080 – preventing manual entry in Catalog PII Edit section. Previously it was possible to add ‘pii’ and ‘classification’ properties incorrectly (e.g. in capital letters). It created confusion as they appeared similar in the UI, but the system didn’t recognize them. The problem has been resolved. Now, if the user types ‘pii’ and ‘classification’ in any case (capital or camel case), it is corrected to small letters.
- Ticket #42148 – Fabric Instance Group - Σ No. of instances not working. The problem has been fixed.
- Ticket #42733 – MTable selection is not sorted. The problem has been resolved. The MTables are now arranged in a lexicographical order which will make it easier to search.
- Ticket #43072 – New FabricSetRead actor not working when LU is provided on global name. The problem has been fixed.
- Ticket #43557 – Catalog run on Cassandra is identifying as BYTES instead of Timestamp. The problem has been fixed.



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- Ticket #43643 – Can't insert string to a TEXT field. The problem has been fixed.